# MSG No. 149.22 $P3121'$ [ Type II, trigonal ]

Table 1: Wyckoff site: `1a`, site symmetry: `3.21'`

| No. | position | mapping |
|---|---|---|
| 1 | $[0, 0, 0]$ | `[1,2,3,4,5,6,7,8,9,10,11,12]` |

Table 2: Wyckoff site: `1b`, site symmetry: `3.21'`

| No. | position | mapping |
|---|---|---|
| 1 | $\left[0, 0, \frac{1}{2}\right]$ | `[1,2,3,4,5,6,7,8,9,10,11,12]` |

Table 3: Wyckoff site: `1c`, site symmetry: `3.21'`

| No. | position | mapping |
|---|---|---|
| 1 | $\left[\frac{1}{3}, \frac{2}{3}, 0\right]$ | `[1,2,3,4,5,6,7,8,9,10,11,12]` |

Table 4: Wyckoff site: `1d`, site symmetry: `3.21'`

| No. | position | mapping |
|---|---|---|
| 1 | $\left[\frac{1}{3}, \frac{2}{3}, \frac{1}{2}\right]$ | `[1,2,3,4,5,6,7,8,9,10,11,12]` |

Table 5: Wyckoff site: `1e`, site symmetry: `3.21'`

| No. | position | mapping |
|---|---|---|
| 1 | $\left[\frac{2}{3}, \frac{1}{3}, 0\right]$ | `[1,2,3,4,5,6,7,8,9,10,11,12]` |

Table 6: Wyckoff site: `1f`, site symmetry: `3.21'`

| No. | position | mapping |
|---|---|---|
| 1 | $\left[\frac{2}{3}, \frac{1}{3}, \frac{1}{2}\right]$ | `[1,2,3,4,5,6,7,8,9,10,11,12]` |

Table 7: Wyckoff site: `2g`, site symmetry: `3..1'`

| No. | position | mapping |
|---|---|---|
| 1 | $[0, 0, z]$ | `[1,2,3,7,8,9]` |
| 2 | $[0, 0, -z]$ | `[4,5,6,10,11,12]` |

Table 8: Wyckoff site: 2h, site symmetry: 3..1'

| No. | position | mapping |
|---|---|---|
| 1 | $\left[\frac{1}{3},\ \frac{2}{3},\ z\right]$ | [1,2,3,7,8,9] |
| 2 | $\left[\frac{1}{3},\ \frac{2}{3},\ -z\right]$ | [4,5,6,10,11,12] |

Table 9: Wyckoff site: 2i, site symmetry: 3..1'

| No. | position | mapping |
|---|---|---|
| 1 | $\left[\frac{2}{3},\ \frac{1}{3},\ z\right]$ | [1,2,3,7,8,9] |
| 2 | $\left[\frac{2}{3},\ \frac{1}{3},\ -z\right]$ | [4,5,6,10,11,12] |

Table 10: Wyckoff site: 3j, site symmetry: ..21'

| No. | position | mapping |
|---|---|---|
| 1 | $[x,\ -x,\ 0]$ | [1,6,7,12] |
| 2 | $[x,\ 2x,\ 0]$ | [2,4,8,10] |
| 3 | $[-2x,\ -x,\ 0]$ | [3,5,9,11] |

Table 11: Wyckoff site: 3k, site symmetry: ..21'

| No. | position | mapping |
|---|---|---|
| 1 | $\left[x,\ -x,\ \frac{1}{2}\right]$ | [1,6,7,12] |
| 2 | $\left[x,\ 2x,\ \frac{1}{2}\right]$ | [2,4,8,10] |
| 3 | $\left[-2x,\ -x,\ \frac{1}{2}\right]$ | [3,5,9,11] |

Table 12: Wyckoff site: 6l, site symmetry: 11'

| No. | position | mapping |
|---|---|---|
| 1 | $[x,\ y,\ z]$ | [1,7] |
| 2 | $[-y,\ x-y,\ z]$ | [2,8] |
| 3 | $[-x+y,\ -x,\ z]$ | [3,9] |
| 4 | $[x,\ x-y,\ -z]$ | [4,10] |
| 5 | $[-x+y,\ y,\ -z]$ | [5,11] |
| 6 | $[-y,\ -x,\ -z]$ | [6,12] |